

Dual Iron-Catalyzed Selective Aerobic Lactonization of 1,6-Diols

Sheng Jiang,[§] Yibo Yu,[§] Yankai Huang, Jueqi Dong, Hui Qian,^{*} and Shengming Ma^{*}Cite This: *ACS Catal.* 2025, 15, 13704–13711

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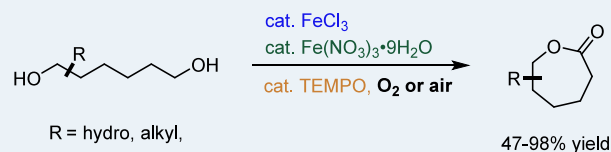
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ABSTRACT: Selectivity control is a very challenging topic in catalytic oxidations of alcohols using oxygen as an oxidant. Of particular interest is the catalytic aerobic lactonization of diols; due to the selectivity issue referring to the two hydroxy groups, the reaction afforded a mixture of isomeric products. Herein, we report an efficient catalytic recipe of easily available earth-abundant metal catalysts $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and FeCl_3 together with TEMPO for the highly selective aerobic oxidative lactonization of 1,6-diols, affording a variety of ϵ -caprolactones. The reaction was conducted under mild conditions with an O_2 balloon or under ambient air enjoying high chemo- and regioselectivity for the oxidative lactonization of non-symmetric diols by the subtle recognition of the steric environment of the two hydroxy groups. Furthermore, reduction and selective aerobic oxidative lactonization constitute a formal carbonyl walking over the ϵ -caprolactones. Mechanistic experiments reveal a process with a hemiacetal as the intermediate. This protocol unlocks an efficient pathway to derive monomers for polycaprolactones.

KEYWORDS: caprolactone, diols, iron catalysis, aerobic oxidative lactonization, polycaprolactone



INTRODUCTION

Lactones are very common structural units in natural products and essential building blocks in organic synthesis.^{1,2} Especially, ϵ -caprolactones are indispensable monomers in the production of polycaprolactones (PCLs),³ which have been widely applied in drug-delivery systems and medical devices such as sutures and wound dressings (Scheme 1a).⁴ On this basis, the incorporation of functional groups into the PCL can help tune the hydrophilicity, bioadhesion, and biodegradation rate.^{5,6} In principle, there are two approaches to the synthesis of ϵ -caprolactones: the Baeyer–Villiger oxidation of symmetric cyclohexanones^{7,8} and oxidative lactonization of 1,6-diols (Scheme 1b).⁹ 2- and 4-Functionalized caprolactone monomers are the most prevailing.^{10,11} Thus, highly selective aerobic preparation of differently substituted non-symmetric caprolactones is of current interest and high demand. The traditional approaches utilized at least stoichiometric amounts of oxidants.^{12–28} Thus, aerobic oxidation using oxygen would, in principle, provide a highly competitive alternative for environmental concerns. However, up to now even the formation of the parent ϵ -caprolactone from 1,6-hexandiol is challenging because of the unfavorable enthalpic and entropic effects of the ring.^{4,29–34} Remarkable catalytic recipes have been developed by Shimizu,²⁷ Kara,³⁵ Liu and Zhu,³⁰ Ji, Bäckvall,³⁷ Li,³⁸ Gawande and Zboril,³⁹ Zhou,⁴⁰ and their coworkers, affording the parent ϵ -caprolactone from the parent 1,6-hexandiol. In addition, aerobic oxidative lactonization of non-symmetric primary 1,6-diols with steric discrepancy remains elusive since all of the known catalytic recipes forming a mixture of isomeric lactones^{34,36,37,40,41} and other oxidative byproducts including ol-als and diols.^{18,21,24} Inspired by our early study on the $\text{Fe}(\text{NO}_3)_3$ -TEMPO-cocatalyzed aerobic

oxidation of primary alcohols,^{42–46} we explored the oxidation of the parent 1,6-hexandiol under our previously reported conditions combining $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ /TEMPO/ KCl ⁴³ and observed that it *disappointingly* afforded the desired ϵ -caprolactone in merely 1% yield by NMR, together with 9% NMR yield of 1,6-hexandiol (Scheme 1c). Here, we wish to report the development of a new set of catalytic recipes for the highly chemo- and regioselective aerobic oxidative lactonization of 1,6-diols under mild reaction conditions, while successfully addressing the challenge of regioselectivity (Scheme 1d).

RESULTS AND DISCUSSION

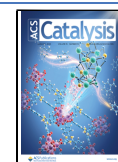
As already stated, our group reported the iron nitrate-catalyzed approaches for aerobic oxidation of alcohols by applying different cocatalysts.^{42–46} However, only a trace of ϵ -caprolactone **2a** and 11% yield of 1,6-hexandiol **3a** was formed in the reaction of 1,6-hexandiol **1a** upon using $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, TEMPO, and NaCl (Table 1, entry 1) as catalysts. These results prompted us to identify new cocatalysts. To our delight, after comprehensive screening, we observed that the aerobic oxidation of 1,6-hexandiol **1a** with BiCl_3 as the cocatalyst afforded a mixture of ϵ -caprolactone **2a** (53% yield),

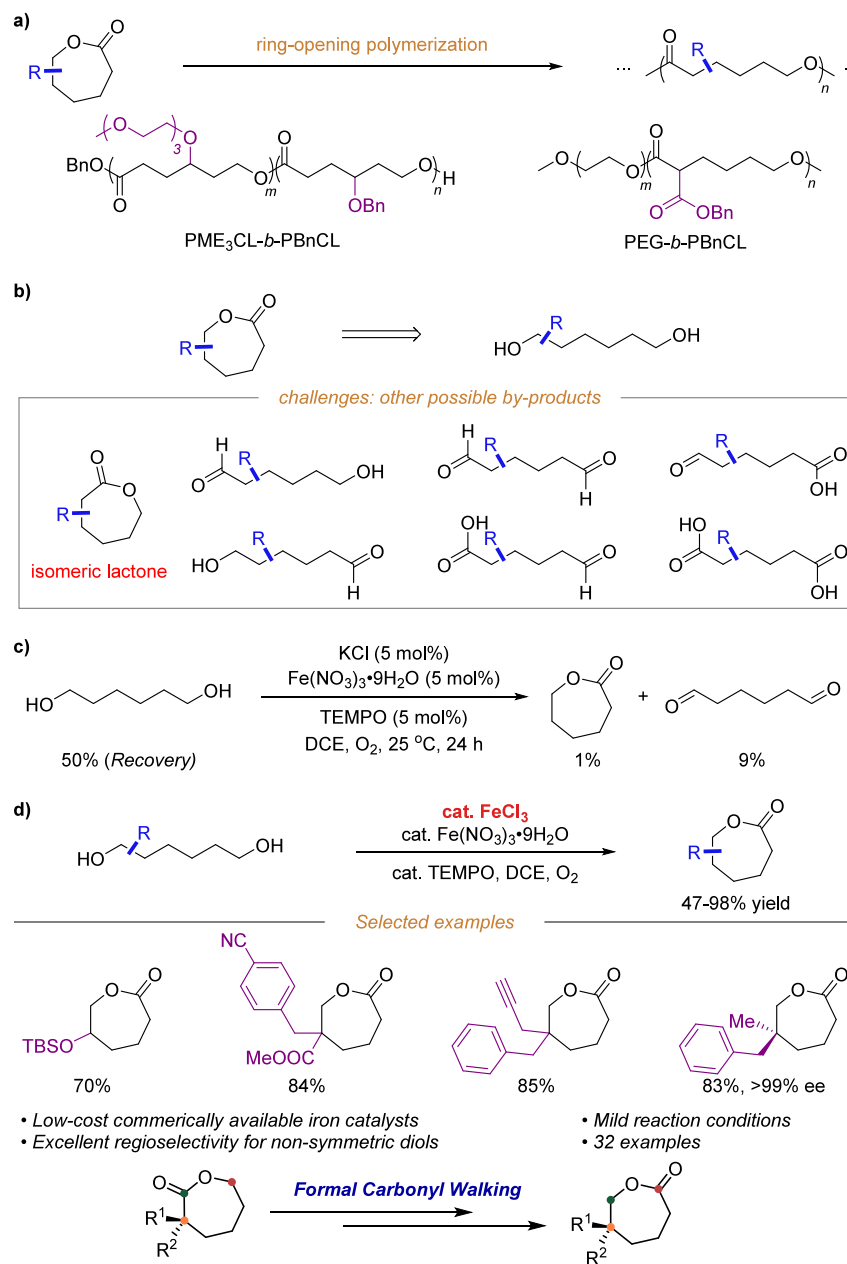
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Scheme 1. Importance and Approaches to Non-symmetric ϵ -Caprolactones^a

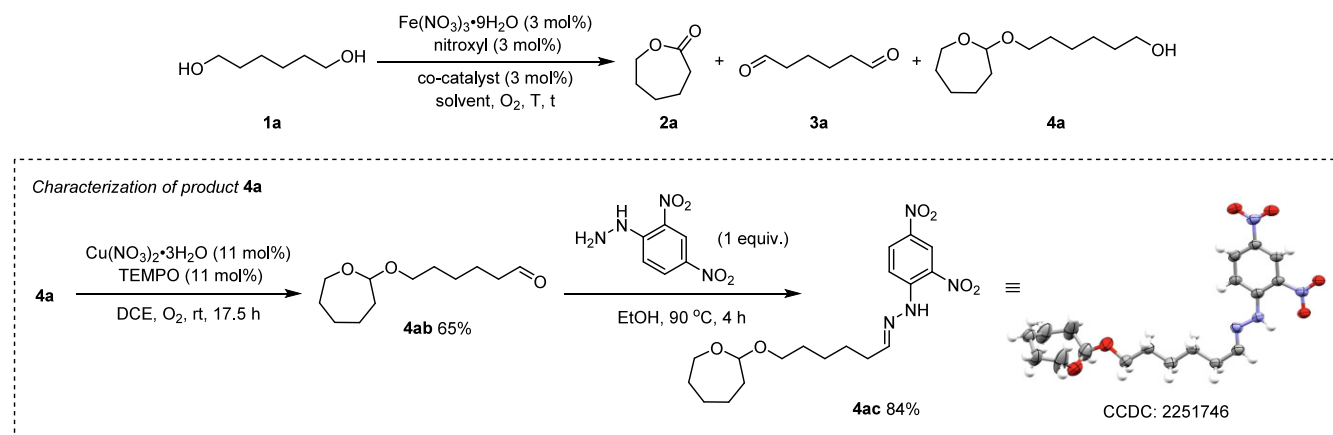
^a(a) ϵ -Caprolactone-based polyesters. (b) Oxidative lactonization of 1,6-diols for non-symmetric ϵ -caprolactones and known byproducts. (c) Preliminary study on aerobic oxidative lactonization of 1,6-hexandiol. (d) This work: dual iron-catalyzed highly selective aerobic lactonization.

dial **3a** (18% yield), and bimolecular alcohol with an acetal unit **4a** (22% yield) (Table 1, entry 2).

When 15 mol % of NaCl, which matched the stoichiometry of chloride in the used BiCl₃, was applied, 4% yield of ϵ -caprolactone **2a**, 14% yield of **3a**, and 24% yield of **4a** were formed with 41% recovery of **1a** (Table 1, entry 3). These results excluded the possibility that chloride ions alone could improve the catalytic efficiency. The structure of the bimolecular product **4a** was firmly established by an X-ray crystallographic study of **4ac** after copper-catalyzed aerobic oxidation⁴⁷ and the subsequent reaction with 2,4-dinitrophenylhydrazine. When the loading of catalysts was reduced to 3 mol % each of Fe(NO₃)₃·9H₂O, TEMPO, and BiCl₃, ϵ -caprolactone **2a** was obtained in 70% yield with 6% yield of **4a** (Table 1, entry 4). We then screened various TEMPO

derivatives with no better results (Table 1, entries 5–7). The solvent screening indicated that the oxidation proceeded almost completely in chlorinated solvents such as DCM, CHCl₃, and PhCl with a much poorer selectivity (Table 1, entries 8–10). Subtle adjustments of the loading of TEMPO to 2 mol % and running the reaction at 35 °C for 24 h afforded the product in a similar yield (Table 1, entry 12).

We next executed an intensive optimization of the cocatalysts for Fe(NO₃)₃·9H₂O-TEMPO combination. The alkaline metal salt MgCl₂ was not suitable (Table 1, entry 13). To our surprise, AlCl₃ and FeCl₃ performed well, forming ϵ -caprolactone **2a** in ~ 70% yield (Table 1, entries 15 and 16). We also conducted the reaction with 9 mol % NaCl instead of 3 mol % FeCl₃. However, only a trace of ϵ -caprolactone **2a** was obtained (Table 1, entry 17). In view of the cost and

Table 1. Optimization of Reaction Conditions and Characterization of the Product 4a^a

entry	nitroxyl (mol %)	cocatalyst	solvent	T (°C)	t (h)	recovery (%)	NMR yield (%)		
							2a	3a	4a
1 ^b	TEMPO (5)	NaCl	DCE	25	24	46	1	11	0
2 ^b	TEMPO (5)	BiCl ₃	DCE	25	24	3	53	18	22
3 ^c	TEMPO (5)	NaCl	DCE	25	24	41	4	14	24
4	TEMPO (3)	BiCl ₃	DCE	30	36	0	70	13	6
5	4-OH-TEMPO (3)	BiCl ₃	DCE	30	36	3	52	19	30
6	4-NH ₂ -TEMPO (3)	BiCl ₃	DCE	30	36	19	5	6	62
7	4-NHAc-TEMPO (3)	BiCl ₃	DCE	30	36	1	65	19	trace
8	TEMPO (3)	BiCl ₃	DCM	30	36	6	29	11	40
9	TEMPO (3)	BiCl ₃	CHCl ₃	30	36	0	57	26	0
10	TEMPO (3)	BiCl ₃	PhCl	30	36	4	20	18	46
11	TEMPO (3)	BiCl ₃	MTBE	30	36	37	3	11	36
12	TEMPO (2)	BiCl ₃	DCE	35	24	3	70	16	2
13	TEMPO (2)	MgCl ₂	DCE	35	24	53	6	9	0
14	TEMPO (2)	InBr ₃	DCE	35	24	27	30	5	24
15	TEMPO (2)	AlCl ₃	DCE	35	24	4	70	18	6
16	TEMPO (2)	FeCl ₃	DCE	35	24	1	72	15	0
17 ^d	TEMPO (2)	NaCl	DCE	35	24	73	1	7	6
18 ^e	TEMPO (4)	FeCl ₃	DCE	35	48	0	64	18	0

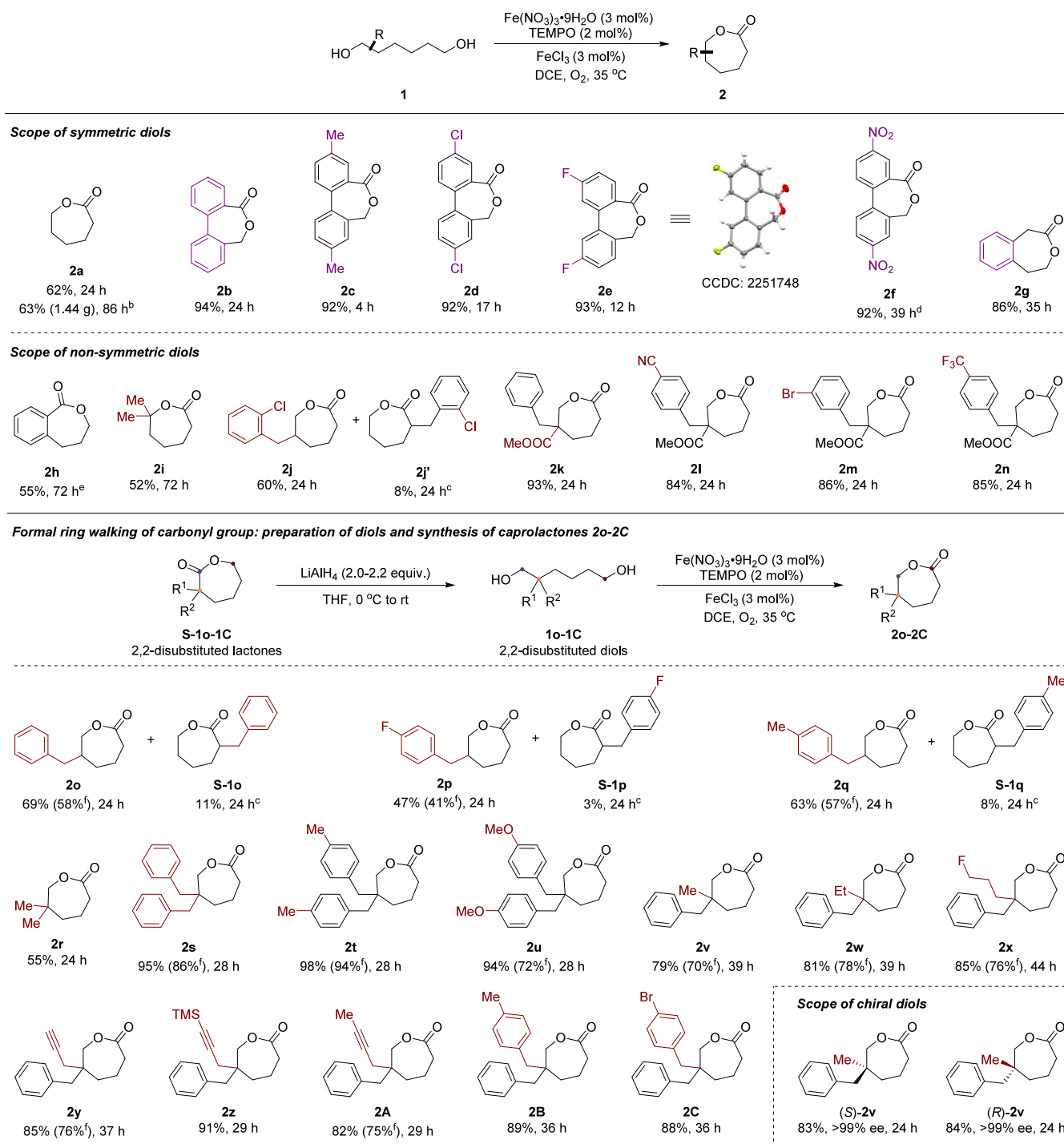
^aReactions were conducted with 1a (1 mmol), Fe(NO₃)₃·9H₂O (3 mol %), cocatalyst (3 mol %), and nitroxyl in solvent (5 mL) with an O₂ balloon unless otherwise noted. NMR yields and recoveries were determined by using CH₂Br₂ as the internal standard. ^bWith 5 mol % each of Fe(NO₃)₃·9H₂O, TEMPO, and cocatalyst. ^cWith 15 mol % NaCl. ^dWith 9 mol % NaCl. ^eAmbient air with 4 mol % each of Fe(NO₃)₃·9H₂O, TEMPO, and FeCl₃.

practicability, we chose FeCl₃ as the optimal cocatalyst. The reaction may also be conducted under ambient air with a 65% yield of 2a (Table 1, entry 18).

With the optimized conditions in hand (Table 1, entry 16), we first examined the scope of symmetric 1,6-diols (Scheme 2). A 20 mmol scale reaction of 1,6-hexanediol 1a afforded 1.44 g (63%) of 2a. Aerobic oxidative lactonization of diols 1b–1f with the biphenyl unit afforded lactones 2b–2f exclusively in excellent yields,^{26,27,34,40,41} and the electron-donating methyl group and electron-withdrawing groups of chloro, fluoro, and nitro at 4,4' or 5,5'-position of biphenyl were compatible. *ortho*-Phenyl-tethered diol 1g could also readily undergo this transformation to deliver lactones 2g.

The selectivity issue arises upon using non-symmetric diols: for *ortho*-phenyl-tethered diol 1h, benzyl alcohol was oxidized to afford the lactone 2h. The reaction of diol 1i with a highly hindered tertiary alcohol unit afforded lactone 2i in 52% yield as expected. Surprisingly, a remarkable selectivity was observed even for oxidative lactonization of 2-monosubstituted diol 1j, affording 5-substituted lactone 2j as the major product (60% yield), while only an 8% yield of the isomeric 2-substituted

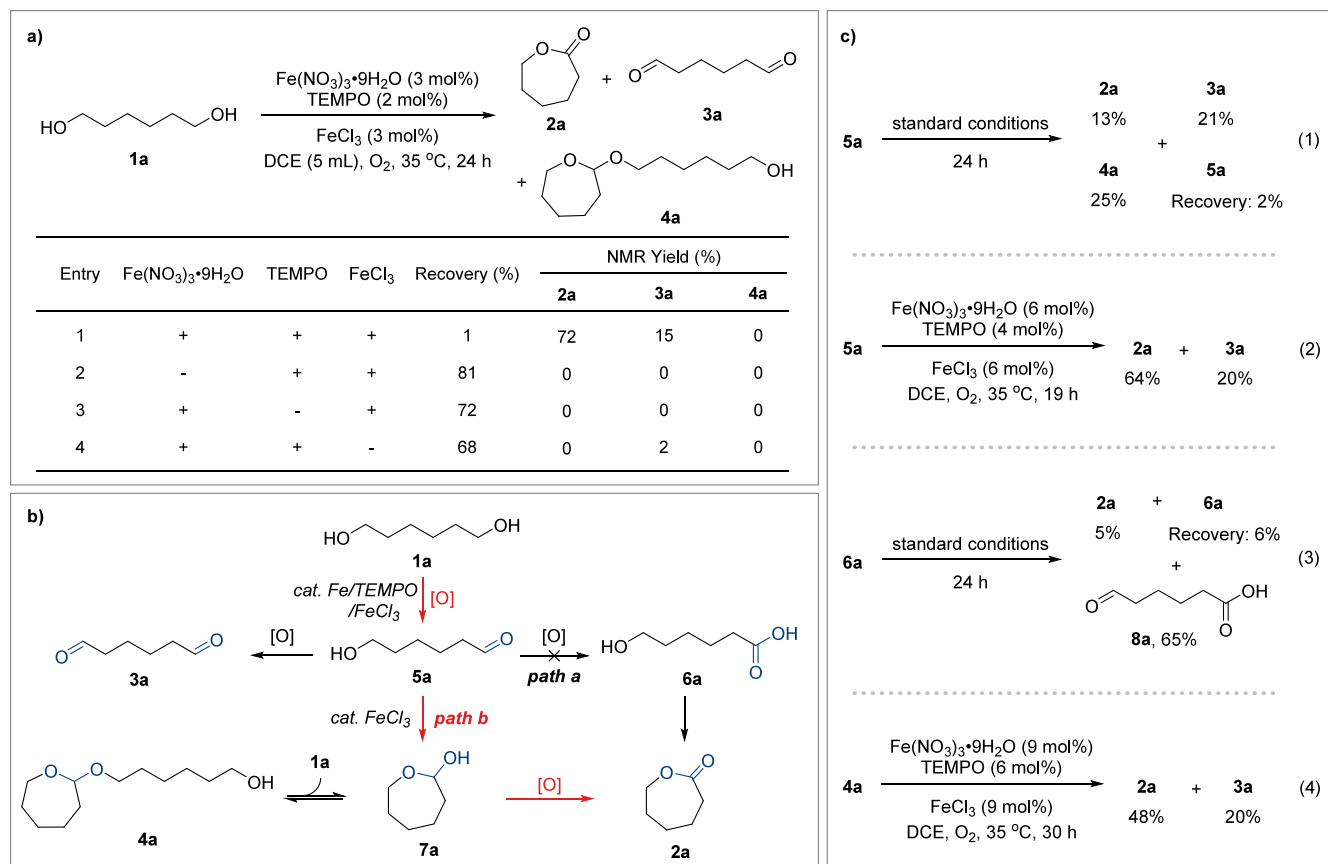
lactone 2j' was formed, indicating a subtle steric preference of the less sterically hindered hydroxy group for this catalytic recipe. In the literature,^{34,36,37,40,41} such diols with a small steric difference for the hydroxy group led to the formation of a mixture of lactone products. Such a steric effect has been further enforced with 2,2-disubstituted diols 1k–1n, delivering the corresponding 5,5-disubstituted lactones 2k–2n exclusively in good to excellent yields. Synthetically versatile ester, cyano, and bromo and potentially useful trifluoromethyl are well tolerated. Since these 2-substituted diols were prepared from the reduction of 2-substituted lactones with LiAlH₄, this transformation to 5-substituted lactones realized a formal two-step carbonyl group ring walking over the caprolactones, affording 5-monosubstituted lactones 2o–2q in high regioselectivity and 5,5-disubstituted lactones 2r–2c exclusively. Functional groups such as methoxy (2u), terminal C–C triple bond (2y), non-terminal C–C triple bond (2z and 2A), benzyl (2B), and C–Br bond (2C) are all compatible. Notably, chiral diols (S)-1v and (R)-1v could be smoothly transformed to optically active lactones (S)-2v and (R)-2v without racemization.

Scheme 2. Scope of Aerobic Oxidative Lactonization of 1,6-Diols^a

^aReaction conditions: **1** (1.0 mmol), Fe(NO₃)₃·9H₂O (3 mol %), TEMPO (2 mol %), FeCl₃ (3 mol %) in DCE (5 mL) at 35 °C with an O₂ balloon unless otherwise noted. ^b20 mmol scale reaction. ^cNMR yields were determined by using CH₂Br₂ as the internal standard. ^d45 °C. ^e25 °C. ^fCombined yield of reduction and oxidative lactonization.

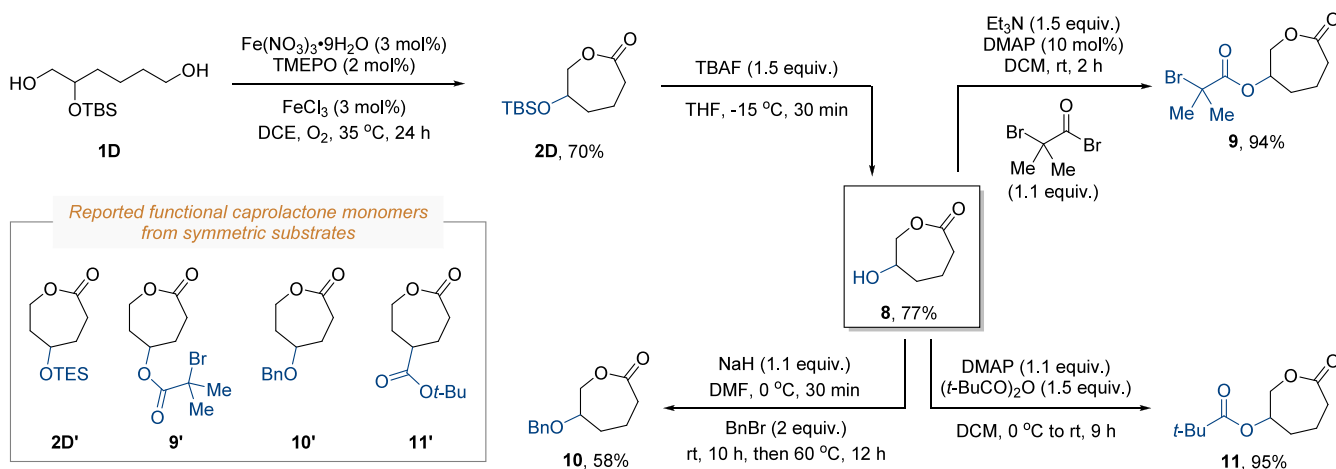
To demonstrate the necessity of each cocatalyst, some control experiments were carried out. No ϵ -caprolactone **2a** was formed in the absence of TEMPO or Fe(NO₃)₃·9H₂O (Scheme 3a, entries 2 and 3), and only a 2% yield of hexandial **3a** was formed without FeCl₃ (Scheme 3a, entry 4), proving that each catalyst was indispensable in this oxidative lactonization. Thus, two plausible reaction pathways were proposed, as shown in Scheme 3b. We assumed that the initial

oxidation of 1,6-hexandiol **1a** would give 6-hydroxyhexanal **5a**, which may be transformed to 6-hydroxyhexanoic acid **6a** and then esterified to furnish ϵ -caprolactone **2a** (Scheme 3b, path a). Alternatively, 6-hydroxyhexanal **5a** may also undergo intramolecular 1,2-addition to afford hemiacetal **7a** with the assistance of FeCl₃, serving as the Lewis acid to activate the carbonyl unit (path b), which would be further oxidized to form ϵ -caprolactone **2a**. Surprisingly, the treatment of 6-

Scheme 3. Mechanistic Studies^a

^a(a) Control experiments: **1a** (1 mmol), with or without Fe(NO₃)₃·9H₂O (3 mol %), with or without TEMPO (2 mol %), with or without FeCl₃ (3 mol %) in DCE (5 mL) at 35 °C with an O₂ balloon for 24 h. NMR yields and recoveries were determined by using CH₂Br₂ as the internal standard. (b) Plausible reaction pathways. (c) Supporting experiments: reactions of the possible intermediates. NMR yield was determined by using CH₂Br₂ as an internal standard.

Scheme 4. Syntheses of Diversified Caprolactones



hydroxyhexanal **5a**⁴⁸ under the standard conditions afforded ϵ -caprolactone **2a** in only 13% yield, with 21% of dial **3a**, 25% of acetal-alcohol **4a**, and 2% recovery of **5a** (Scheme 3c, eq 1). We speculated that the in situ loading of catalysts with respect to the intermediate **5a** under the standard conditions was much higher than that in eq 1 of Scheme 3c. Thus, the amount of each cocatalyst was increased up to 6 mol % of Fe(NO₃)₃·9H₂O, 4 mol % of TEMPO, and 6 mol % of FeCl₃ (Scheme 3c,

eq 2), and the reaction of **5a** smoothly gave **2a** in 64% yield with 20% of dial **3a**. On the other hand, 6-hydroxyhexanoic acid **6a** was prepared⁴⁹ on purpose and treated under the standard conditions (Scheme 3c, eq 3). However, only a 5% yield of lactone **2a** was observed with 65% yield of 5-formylvaleric acid **8a**. Therefore, **6a** could at least be ruled out as the major intermediate here. In addition, acetal **4a** could also be converted to **2a** and hexandial **3a** with a catalyst

loading close to the reaction conditions (Scheme 3c, eq 4), indicating the easy interconversion between hemiacetals 7a and 4a. These results demonstrated that the oxidative lactonization proceeded through path b with hemiacetal 7a as the key intermediate.

Functionalized PCLs have been synthesized by ring-opening polymerization of corresponding caprolactone monomers.³ Given that the reported caprolactone monomers feature 2- and 4-substituted functional groups,^{10,11} we attempted to make full use of the current approach for new caprolactones with untapped potential. Fortunately, 5-substituted caprolactone 2D could be obtained through the oxidative lactonization of diol 1D with an excellent selectivity (Scheme 4). The deprotection of 2D was successfully achieved to form the non-symmetric caprolactone 8 with a free hydroxy group. Further functionalization of 8 afforded compound carboxylate 9 with 2-bromoisobutryl bromide, benzyl ether 10 with benzyl bromide, and carboxylate 11 with pivalic anhydride, respectively. It is worth noting that the symmetric homologue of 2D (compound 2D')⁵⁰ and the isomers of 9–11 (compounds 9'–11')^{51–53} have been applied as the building blocks of PCL. We expect that our strategy may pave the way for the synthesis of new caprolactone monomers and functionalized polymers.

CONCLUSIONS

In summary, the catalytic recipe of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ /TEMPO/ FeCl_3 was developed for the highly selective synthesis of ϵ -caprolactones via the aerobic oxidative lactonization of 1,6-diols with an O_2 balloon or under ambient air. The presence of an additional catalytic component of FeCl_3 is critical for the reactivity and regioselectivity control. We reasoned that the addition of FeCl_3 may have formed a bulkier catalyst, differentiating two types of sterically different hydroxyl groups. Combining the reduction with LiAlH_4 and the current oxidation, formal carbonyl walking from 2-mono- or 2,2'-disubstituted caprolactones to 5-mono- or 5,5'-disubstituted caprolactones was realized. Non-symmetric new caprolactones may manifest the potential of this method in materials science.

METHODS

General Procedure for Iron-Catalyzed Selective Aerobic Lactonization of 1,6-Diols. To a Schlenk tube were added FeCl_3 (5.1 mg, 0.03 mmol), TEMPO (0.02 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (0.03 mmol), 1,6-diol 1 (1.0 mmol), and DCE (5 mL). The resulting mixture was then stirred at 35 °C under the atmosphere of O_2 from a balloon until completion of the reaction, as monitored by TLC, filtered through a short column of silica gel eluted with ethyl acetate (25 mL $\times$ 3), and concentrated. The resulting mixture was dissolved in 5 mL of MeCN, stirred with a saturated aqueous solution of NaHSO_3 (10 mL) for 2 h, diluted with 10 mL of H_2O , extracted with ethyl acetate (15 mL $\times$ 3), and dried over anhydrous Na_2SO_4 . After filtration and evaporation, the residue was purified by column chromatography on silica gel to afford the corresponding lactone 2.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.5c03444>.

Detailed experimental procedures, characterization of new compounds (PDF)

AUTHOR INFORMATION

Corresponding Authors

Hui Qian – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, China; orcid.org/0000-0002-5606-9213; Email: qian_hui@fudan.edu.cn

Shengming Ma – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, China; State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai 200032, China; orcid.org/0000-0002-2866-2431; Email: masm@sioc.ac.cn

Authors

Sheng Jiang – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, China

Yibo Yu – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, China

Yankai Huang – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, China

Jueqi Dong – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, China

Complete contact information is available at:

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Author Contributions

[§]S.J. and Y.Y. contributed equally.

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